MILITARY SPECIFICATION

CUVETTE, SPECTROPHOTOMETER

This limited coordination Military Specification has been prepared by the Defense Personnel Support Center based upon currently available technical information. It is subject to modification. However, pending its promulgation as a coordinated Military Specification, it may be used in procurement.

1. SCOPE

1.1 This specification covers round, evacuation type, glass cuvettes for use with spectrophotometers (see 6.1.1) in laboratory procedures.

1.2 Classification.

1.2.1 Type and sizes shall be as follows:

Type I. With external evacuation

Size 1 - 10 x 95 mm

Size 2 - 12 x 95 mm

Type II. With internal evacuation

Size 1 - 14 x 106 mm

Size 2 - 19 x 120 mm

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

FSC 6640

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SPECIFICATIONS

Federal

DD-G-541	Glass, Laboratory.
MMM-A-250	Adhesive, Water-Resistant (For Sealing Fiberboard Boxes).
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water-Resistant (For Carton Sealing).

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by supplies in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM Designation C162	Standard Definitions of Terms Relating to Glass and Glass Productions (including Tentative Revisions).

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ASTM Designation C148 — Polariscope Examination of Glass Containers.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Stree, Philadelphia, Pa. 19103).

3. REQUIREMENTS

3.1 Material.

3.1.1 Glass. Glass for cuvettes and component glass parts shall be of borosilicate glass and shall be Type I, Class A of DD-G-541b with refractive index of 1.47.

3.1.1.1 Annealing. Glass shall be annealed so that ring strain shall be no greater than that of temper no. 1, when tested as specified in 4.4.1. Any longitudinal strain, if present, shall be faint and highly diffused.

3.1.2 Plastic.

3.1.2.1 Tubing. Large and small diameter tubing shall be fabricated of a suitable polyvinyl chloride resin.

3.1.2.2 Reducing connector. Reducing connector shall be made of polypropylene resin.

3.2 Design. Cuvettes shall be fabricated of borosilicate glass (see 3.1.1) and shall be in accordance with type I, figure 1 or type II, figure 2. Cuvettes shall consist of a cylindrical tube with funnel shaped flange, an evacuation tube and an egress tube with a vertical tube segment. Each cuvette shall be furnished with a drainage tubing assembly.

3.2.1 Cylindrical tube. Cuvette tube shall be a cylindrical, clear, borosilicate glass tubing (see 3.1.1) symmetrically round in shape and shall be as shown in figures 1 and 2. Opening at bottom of type I cuvettes shall be centrally positioned and bottom of type II tubes shall be rounded to facilitate complete evacuation of the contained fluid (see 3.4). Mouth end of tubes shall be formed into a symmetrical, funnel shaped opening to facilitate pouring-in of fluid reagents in routine use. Funnel rim shall be beaded and shall be smooth and perpendicular to the axis of the tube.

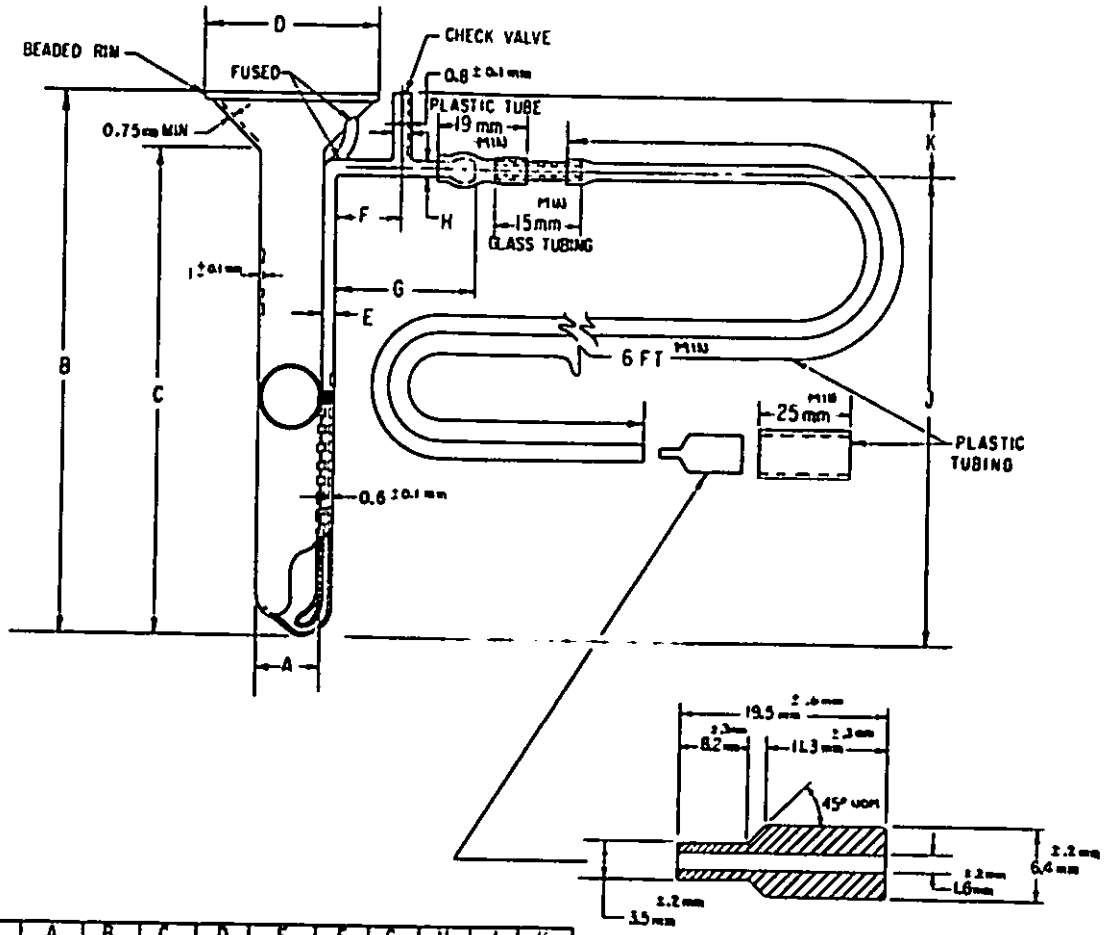
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3.2.1.1 Evacuation tubes. Evacuation tubes in the cuvettes shall be in close proximity to the tube wall as shown in figures 1 and 2. The evacuation tube at the upper end shall connect to the egress tube so that the lumen of both tubes shall be smooth and without constriction. Evacuation tube in type I cuvettes shall have a rounded contour at the bottom without a sharp bend so as to preclude any constriction of the lumen. A piece of glass rod, having the same characteristics as the tube material, shall bond the horizontal arm to the funnel so as to afford adequate support strength and rigidity to the evacuation tube and the egress tube (see 3.2.1.2). The opening of the evacuation tube in type II cuvettes shall be beaded and shall be fused centrally to the inside bottom of the tube. At the upper end, the evacuation tube shall connect to the egress tube so as to effect a sound structural fusion to the cuvette body without a break in the continuity of the tube wall.

3.2.1.2 Egress tube. The egress tube (inverted "T" assembly), shall consist of a horizontal arm and a touch valve control (check valve) and shall connect perpendicularly to the evacuation tube just below the funnel. The check valve shall be fused at right angle to the horizontal arm. Fused junctions shall be sound and smooth and shall be effected without constriction of the lumen. Opening of the check valve shall be cut perpendicular to the axis of the tubing and shall be evenly and smoothly fire polished.

3.2.2 Drainage tubing assembly. The drainage tubing assembly shall be furnished with each cuvette and shall be as shown in figures 1 and 2 to effect evacuation of the sample solution from the cuvette. The small glass tubing in the drainage assembly shall connect to the horizontal arm by a small length of plastic tubing so as to obviate the hazard of damage to the delicate tube segments when detaching or replacing the cuvette. This small plastic tubing shall have nominal dimensions of 0.12 inch ID and 0.025 inch wall thickness. A six foot length of this size plastic tubing shall connect to one end of the glass tubing and to the mating end of the plastic reduction connector. A small length of plastic tubing having nominal dimensions of 1/4 inch ID and 1/16 inch wall thickness shall terminate the drainage tubing assembly and connect to a water filter pump (aspirator, see 6.1.2).

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PLASTIC REDUCTION CONNECTOR

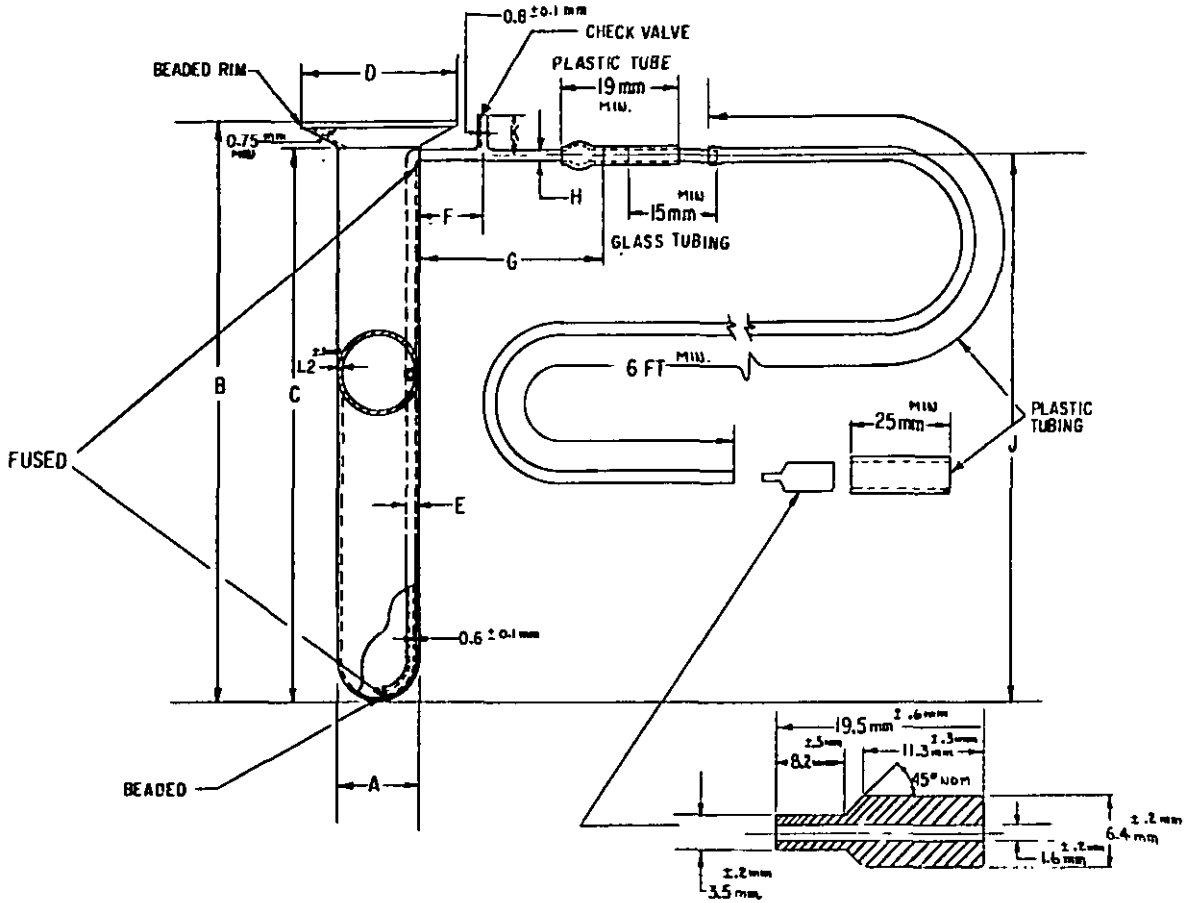
TYPE I	A ±0.2 mm	B ±2 mm	C ±2 mm	D ±2 mm	E ±0.2 mm	F ±1 mm	G ±1 mm	H ±0.3 mm	J ±2 mm	K ±1 mm
SIZE 1 6640-145-1301	10	95	86	31	3	12	25	4	83	10
SIZE 2 6640-145-1300	12	95	86	31	3	12	25	4	83	10

ALL DIMENSIONS IN MILLIMETERS

FIGURE 1

CUVETTE SPECTROPHOTOMETER

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PLASTIC REDUCTION CONNECTOR

TYPE II	A ±0.2mm	B ±2mm	C ±2mm	D ±2mm	E ±0.2mm	F ±1mm	G ±1mm	H ±0.5mm	J ±2mm	K ±1mm
SIZE 1 6640-145-1299	14	106	100	31	3	11	35	4	97	8
SIZE 2 6640-926-7776	19	120	113	30	3	12	25	4	110	13

ALL DIMENSIONS IN MILLIMETERS

FIGURE II

CUVETTE SPECTROPHOTOMETER

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Lumen of the drainage tubing shall be clean and free of occluding particles or other foreign matter. Plastic tubing shall have cleanly cut ends perpendicular to the tubing axis and shall be free from cracks, holes and cuts.

3.3 Performance. Cuvette design shall be such that the tube shall be capable of being completely evacuated and the contained test solution so thoroughly drained that the total residual solution shall show a value not greater than 0.01 optical density, when tested as specified in 4.4.2.

3.4 Identification marking. Each cuvette shall be clearly and permanently marked as follows:

Name or registered trademark of the manufacturer
Dimensions as specified in nomenclature
Federal Stock Number

3.5 Instructions. Instructions for use of the specified cuvettes shall be furnished in each unit package. Instructions shall include procedure for use of the cuvettes showing, by simple illustration, the cuvette with drainage tubing assembly properly connected to the water filter pump (aspirator). Hazards likely to be encountered in the normal use of the cuvettes shall be clearly indicated in the instructions.

NOTE: Instructions shall include a brief description of necessary modifications to the B & L spectrophotometer (See 6.1.1) Type II, Size 1, to enable the use of the cuvette with this instrument.

3.6 Workmanship. Workmanship shall be first class throughout. Cuvettes shall be free from cracks, chips, breaks, crizzles, checks, bubbles, striae, scratches, pits, stones, blisters, coloring or other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the

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performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the borosilicate glass, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor. Certificates of quality, supplied by the manufacturer of the plastic materials, which will contain evidence of the identification of polyvinyl chloride, polyethylene and polypropylene, may be furnished in lieu of actual testing.

4.2 Sampling.

4.2.1 Sampling for examination. Sampling for examination shall be conducted in accordance with MIL-STD-105. The inspection levels and acceptable quality levels shall be as indicated in Table I. Unit of product for examination purposes shall be the cuvette with drainage tube assembly.

Table I. Sampling for Examination

Type Defect	Inspection Level	AQL (Percent Defective)
For Visual Examination		
Major Defects	II	2.5
Minor Defects	II	4.0
For dimensional examination	S-2	2.5

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4.2.2 Sampling for tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II, except that the acceptance number shall be zero for all sample sizes.

Table II. Sampling for Tests

Component (unit of product)	Characteristic	Requirement	Test Procedure	Inspection level
Cuvette	Strain	3.1.1.1	4.4.1	S-4
Cuvette with drainage assembly	Performance	3.3	4.4.2	S-4

4.3 Examination. Cuvette with drainage tube assembly shall be examined to determine compliance with all requirements contained in this specification. Unit item shall be examined for defects including, but not limited to, those listed in Table III. Definitions of glass defects shall be in accordance with ASTM Designation C162.

Table III. Classification of Defects

Categories	Defects
Major	
101	Component part missing.
102	Item identification missing, illegible or incomplete.
103	Cuvette not free of cracks, chips, breaks, crizzles and checks.
104	Rim of funnel not perpendicular to the tube axis.
105	Cuvettes not clear or free of color defects.

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Table III. Classification of Defects (cont'd.)

Categories	Defects
Major	
106	Tube not free of sharp edges and protrusions.
107	Cuvettes not free of seeds, stones, pitting, blisters, bubbles and etched marks.
108	Cuvette tubes not symmetrically round.
109	Hole at bottom of type I cuvettes not centrally located.
110	Bottom of type II cuvettes not round.
111	Funnel shaped mouth end not symmetrical.
112	Rim of funnel shaped opening not smooth and perpendicular to the tube axis.
113	Evacuation tubes in type I and II cuvettes, not in close proximity to body wall.
114	Lumen of evacuation tube and egress tube not connected (fused) without constriction.
115	Evacuation tube not secured (fused) to body wall in type II cuvettes.
116	Check valve not perpendicular to the horizontal arm.
117	Egress tube not perpendicular to the evacuation tube.
118	Fused junctions not smooth and sound.
119	Check valve opening not evenly and smoothly firepolished.
120	Evacuation tube and egress tube lumen not free of occluding particles or other foreign matter.

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Table III. Classification of Defects (cont'd.)

Categories	Defects
Major	
121	Lumen of drainage tubing assembly not free of occluding particles or other foreign matter.
122	Opening of the evacuation tube in type II cuvettes, not beaded and not centrally fused to the inside bottom of the tube.
123	Evacuation tube not secured (fused) to wall at upper end of cuvette type II tube.
Minor	
201	Cuvettes not free of scratches.
202	Cuvettes not free of dirt or foreign matter.
203	Cuvettes not free of mars and blemishes.
204	Cuvettes not free of striae.
205	Check valve opening not cut perpendicular to the tubing axis.
206	Ends of plastic tubing not cleanly cut and perpendicular to the length.
207	Plastic tubing not free of cracks, holes and cuts.

4.3.1 Dimensional examination. The cuvettes with drainage tubing assembly shall be examined for defects in dimensions (See Tables in Figures I and II and 3.2.2). Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests.

4.4.1 Strain test. Cuvettes shall be tested in accordance with method A or B of ASTM C148.

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4.4.2 Performance test.

4.4.2.1 Equipment.

a. Test instrument. Spectrophotometers suitable for use with each type cuvette are listed in 6.1.1.

b. Pump. Water filter pump (aspirator, see 6.1.2).

c. Water pressure at faucet shall be such as to provide efficient evacuation.

d. Indicator solution. Prepare a solution of potassium dichromate (CP) in distilled water so as to attain an optical density of 0.5 + 0.05 when measured in a spectrophotometer set at 550 millimicron wavelength (see 6.1.1).

4.4.2.2 Test procedure.

a. Set up sample cuvette unit in spectrophotometer well, using the proper adapter (see 6.1.3) and stabilize cuvette in position by taping drainage tubing (near cuvette) to the instrument.

b. Set instrument at 550 millimicron wavelength.

c. Zero instrument with distilled water and evacuate until gurgling sound is audible.

d. Introduce, in sample cuvette, 5 milliliters (5 ml) of potassium dichromate indicator solution (0.5 $\pm$ 0.05 optical density), note reading and evacuate.

e. Place 5 ml of distilled water in the same cuvette sample and briefly mix with a suitable stirring rod to insure dissolving of any residual test solution. Note reading, indicating residual optical density.

Note: The residual optical density shall not be greater than 0.01 (see 3.3 requirement).

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One cuvette, spectrophotometer, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be packaged in a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class domestic. Box shall be designed with suitable interior fittings or shall contain sufficient cushioning material to insure against shifting of or damage to the contents. Closure shall be adequate to prevent accidental opening.

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5.1.3.2 Intermediate package. Intermediate package shall be a box of appropriate size and design constructed in accordance with PPP-B-636, type CF, class weather-resistant. Boxes shall be closed by taping the top and bottom flaps with 2 inch wide tape conforming to PPP-T-76. Closing of bottom flaps with adhesive conforming to MMM-A-250 or metal stitches is permitted.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant, grade V3. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

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5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown. Marking shall include the legend: "GLASS - HANDLE WITH CARE".

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129. Marking shall include the legend: "GLASS - HANDLE WITH CARE".

6. NOTES

6.1 Intended use.

6.1.1 Cuvettes specified in this document are suitable for use with the appropriate spectrophotometer as indicated below:

Cuvette (FSN)	Used with Spectrophotometer (FSN)
6640-145-1299	6650-782-5274
6640-145-1300 6640-145-1301 6640-926-7776	6650-144-4555 6650-439-8350

6.1.2 Specified cuvettes are suitable for use with the following items:

FSN 6640-438-3000	Filter, Pump, Water, Jet, Lab.
FSN 6650-935-1557	Vacuum Pump, Reciprocating, Power Driven.

6.1.3 Cuvettes specified in this document are suitable for use with the appropriate adapter as indicated below:

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Cuvette (FSN)	Used with Adapter (FSN)
6640-145-1300	6650-976-2507
6640-145-1301	6650-439-8355
6640-926-7776	6650-439-8360

6.2 This specification covers the following items appearing in the Federal Supply Catalog:

Stock No.	Item Identification	Type	Size
6640-145-1299	CUVETTE, SPECTROPHOTOMETER, 14 by 106 mm.	II	1
6640-145-1300	CUVETTE, SPECTROPHOTOMETER, 12 by 95 mm.	I	2
6640-145-1301	CUVETTE, SPECTROPHOTOMETER, 10 by 95 mm.	I	1
6640-926-7776	CUVETTE, SPECTROPHOTOMETER, 19 by 120 mm.	II	2

6.3 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) Size and type required (See 1.2.1).

(c) Selection of applicable levels of packing (See 5.1 and 5.2).

6.4 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet military requirements.

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Custodians:

Army - MD
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Review activities:

Army - MD
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Preparing activity:

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Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

MIL-C-36611(DSA-DM) FSN

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO (If "yes", in what way?)

4. REMARKS (Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)

SUBMITTED BY (Printed or typed name and activity - Optional)

DATE

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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EDITION AUG 66 WILL BE USED UNTIL EXHAUSTED

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